

ShopEZ

-Commerce Web Application

By

Ezhavarasan. B B.E

Introduction

Shopez is a modern e-commerce web application developed to provide users with a seamless and efficient online shopping experience. The application is designed to allow customers to browse products, add items to the cart, place orders, and manage purchases through a secure and user-friendly platform. It also includes an admin management system for handling products, categories, inventory, and order processing.

The primary goal of Shoppez is to simplify the online shopping process by integrating essential e-commerce functionalities into a single platform. The system is developed using a microservices architecture, where different services such as user management, product management, cart management, and order processing work independently while communicating through APIs. This architecture improves scalability, maintainability, and performance of the application.

Shopez is built using modern technologies including Angular for the frontend, ASP.NET Core Web API for backend services, SQL Server for database management, and JWT Authentication for secure user login and authorization. The platform focuses on delivering high performance, clean user interface design, secure transactions, and efficient product management.

This documentation provides a detailed explanation of the Shoppez system, including its architecture, modules, technologies used, implementation process, and future enhancements.

Student Details

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Acknowledgement

I would like to express my sincere gratitude to **Narasimha Sir** for his valuable guidance, continuous support, and encouragement throughout the development of the **Shopez** project. His insights and technical expertise helped me understand key concepts in software development, microservices architecture, API integration, and full-stack development. I am thankful for the opportunity to learn and successfully complete this project under his mentorship.

I would also like to extend my heartfelt thanks to the **upGrad Team** for providing continuous learning opportunities, resources, and support throughout this journey. Their training programs, structured learning environment, and practical exposure have significantly contributed to enhancing my technical knowledge and professional skills.

Finally, I would like to thank everyone who directly or indirectly supported me in completing this project successfully.

Problem Statement

In the current digital era, customers expect a fast, secure, and convenient shopping experience. Traditional shopping methods and poorly managed online systems often face challenges such as limited accessibility, inefficient product management, lack of secure authentication, poor user experience, and difficulty in handling large amounts of product and order data. Many existing systems also struggle with scalability and maintaining smooth communication between different functionalities.

There is a need for a modern e-commerce platform that enables users to browse products, manage shopping carts, place orders securely, and receive a seamless shopping experience through an easy-to-use interface. At the same time, administrators require an efficient system to manage products, categories, inventory, and customer orders without complexity.

To address these challenges, Shopez is developed as a scalable and efficient e-commerce web application using a microservices architecture. The system aims to provide secure user authentication, efficient product management, smooth order processing, and better system maintainability. By separating functionalities into independent services, Shopez improves performance, scalability, and reliability while ensuring a better shopping experience for users and simplified management for administrators.

Objectives of the Project

The main objective of the Shopez project is to develop a secure, scalable, and user-friendly e-commerce web application that provides customers with a seamless online shopping experience and enables administrators to efficiently manage products and orders.

The specific objectives of the project are:

- 1-To provide a user-friendly shopping platform that allows customers to easily browse, search, and purchase products online.
- 2-To implement secure user authentication and authorization using JWT, ensuring safe login, registration, and account management.
- 3-To develop an efficient product management **system** that allows administrators to add, update, delete, and manage product details and categories.
- 4-To enable shopping cart functionality for users to add, update, and remove products before placing an order.
- 5-To implement an order management system that helps users place orders and allows administrators to monitor and manage order details.
- 6-To build the application using microservices architecture for better scalability, maintainability, and independent service management.
- 7-To ensure smooth communication between frontend and backend systems using RESTful APIs.
- 8-To provide a responsive and attractive user interface for better user experience across different devices.
- 9-To maintain proper database management for storing user, product, cart, and order information efficiently.
- 10-To gain practical knowledge and hands-on experience in full-stack web development, API integration, database management, and software architecture through real-world implementation.

Technology Stack

Angular – Used for developing the user interface (UI) and frontend application.

C# – Used for implementing backend business logic and API development.

ASP.NET Core Web API – Used for building backend APIs and microservices.

Microsoft SQL Server – Used for storing and managing application data.

Postman – Used for testing API endpoints.

Git – Used for source code management and version control.

Visual Studio – Used for backend development and API implementation.

Visual Studio Code – Used for frontend development and code editing.

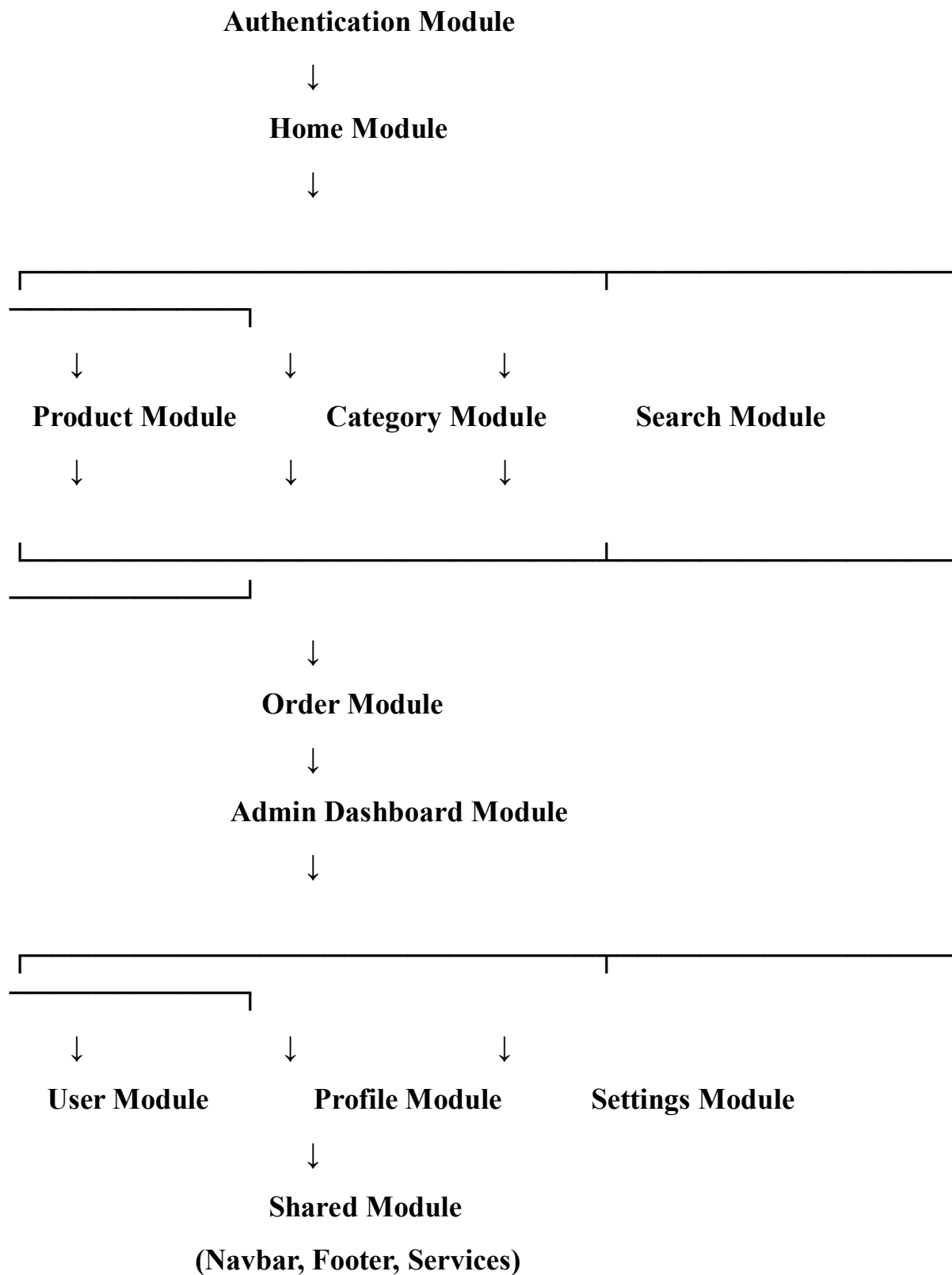
npm – Used for managing Angular packages and dependencies.

Microservices Architecture – Used for developing independent and scalable services.

TypeScript – Used for frontend logic and Angular development.

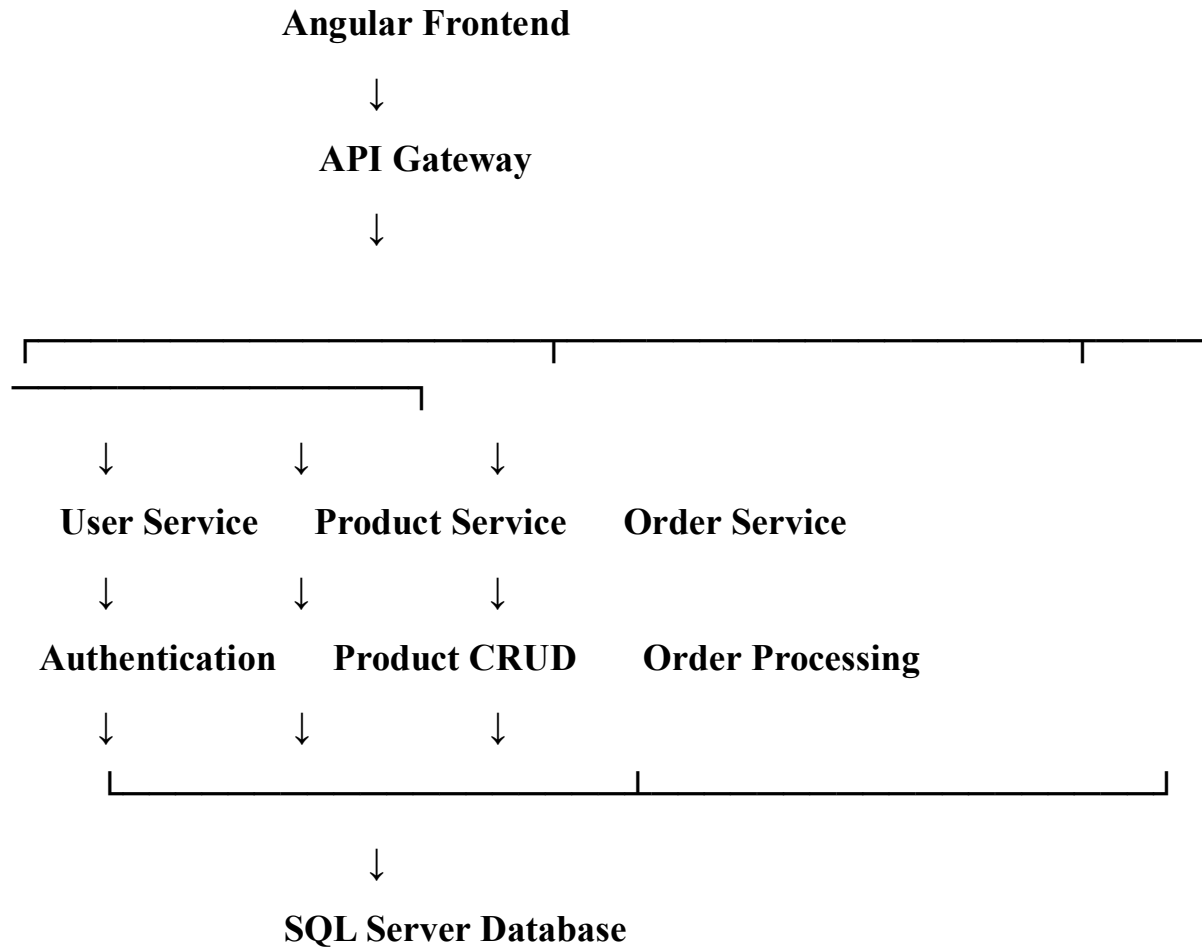
Docker- Used for containerizing and deploying microservices to ensure consistent development and execution across different environments.

FRONTEND MODULES ARCHITECTURE

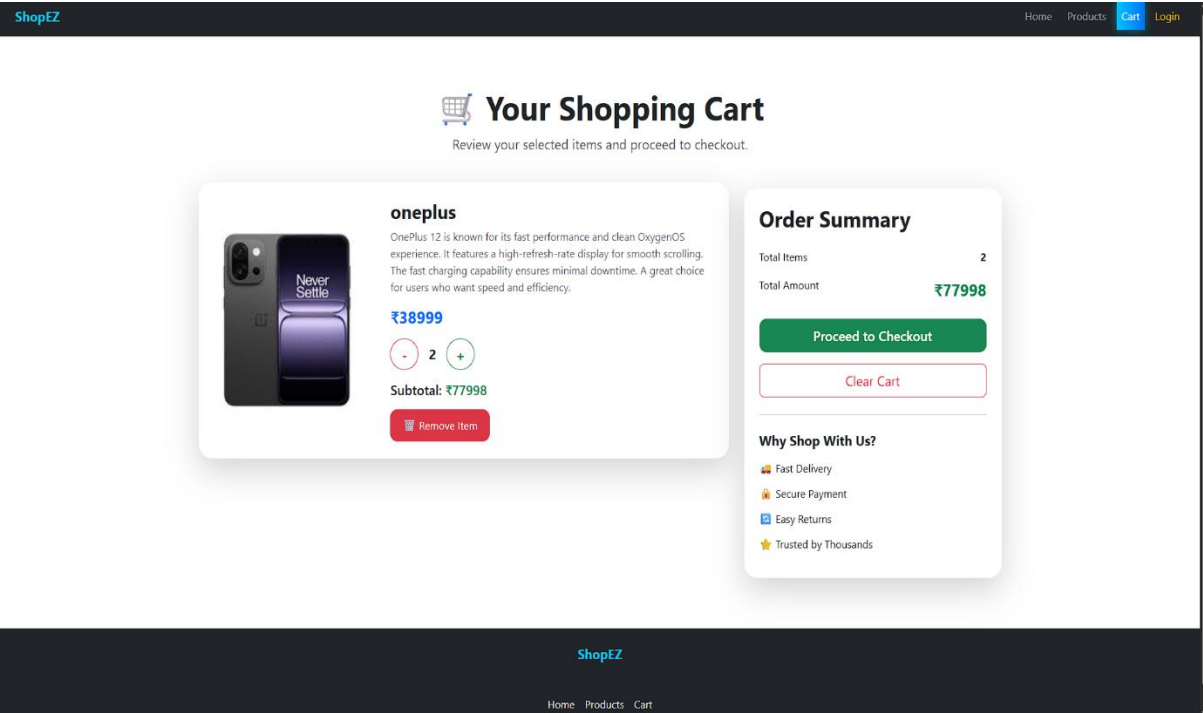
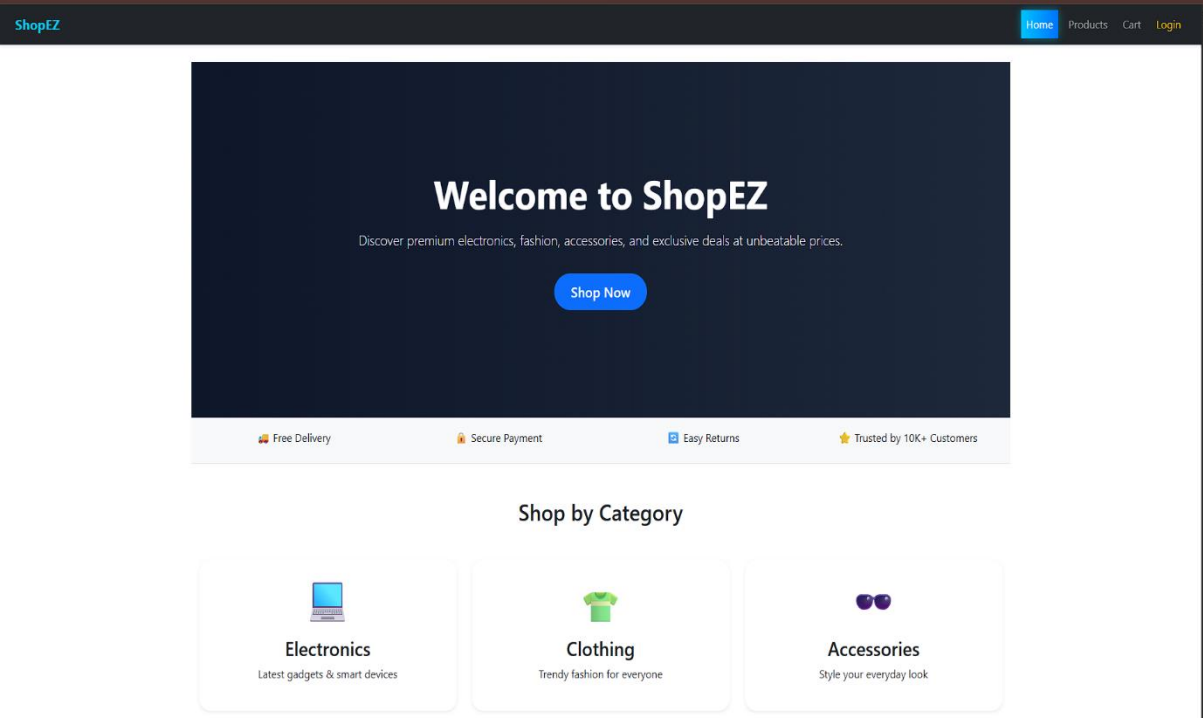


BACKEND MODULES ARCHITECTURE

(Microservices with API Gateway)



Output



Checkout

Complete your order securely and quickly.

Customer Details

Full Name

Enter Full Name

Email Address

Enter Email

Phone Number

Enter Phone Number

Delivery Address

Enter Delivery Address

100% Secure Payments

Fast Delivery Available

Easy Return Policy

Order Summary

oneplus

Quantity: 2

₹77998

Total

₹77998

Place Order

By placing your order you agree to our Terms & Conditions.

ShopEZ

Admin Panel

Products

Users

Orders

Back To Website

Admin Dashboard

Total Products

0

Total Users

0

Total Orders

0

Add Product

Product Name

Enter product name

Price

0

Description

Enter description

Image URL

Enter image URL

Stock

0

Category ID

0

Add Product

Clear

Products List

Name	Description	Price	Stock	Actions
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ShopEZ

Admin Panel

Products

Users

Orders

— Back To Website

Admin Dashboard

Total Products

0

Total Users

0

Total Orders

0

Registered Users		
ID	Name	Email

Testing Output

Karma v 6.4.4 - connected; test: complete; DEBUG

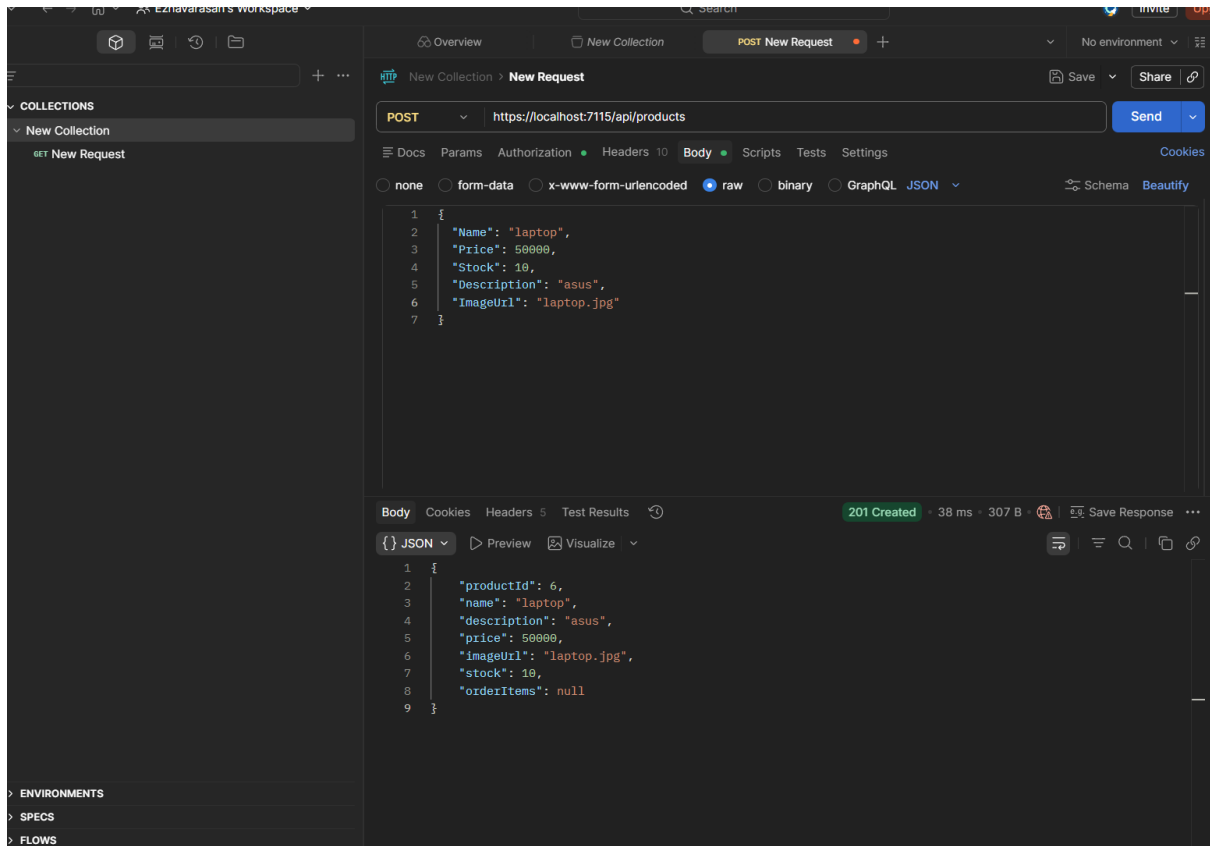
Chrome 148.0.0.0 (Windows 10) is idle

 **Jasmine** 4.6.1 Options

5 specs, 0 failures, randomized with seed 74283 finished in 0.165s

- LoginComponent
 - should create
- AppComponent
 - should create the app
- CartComponent
 - should create
- ProductListComponent
 - should create
- CheckoutComponent
 - should create

Data flow Output

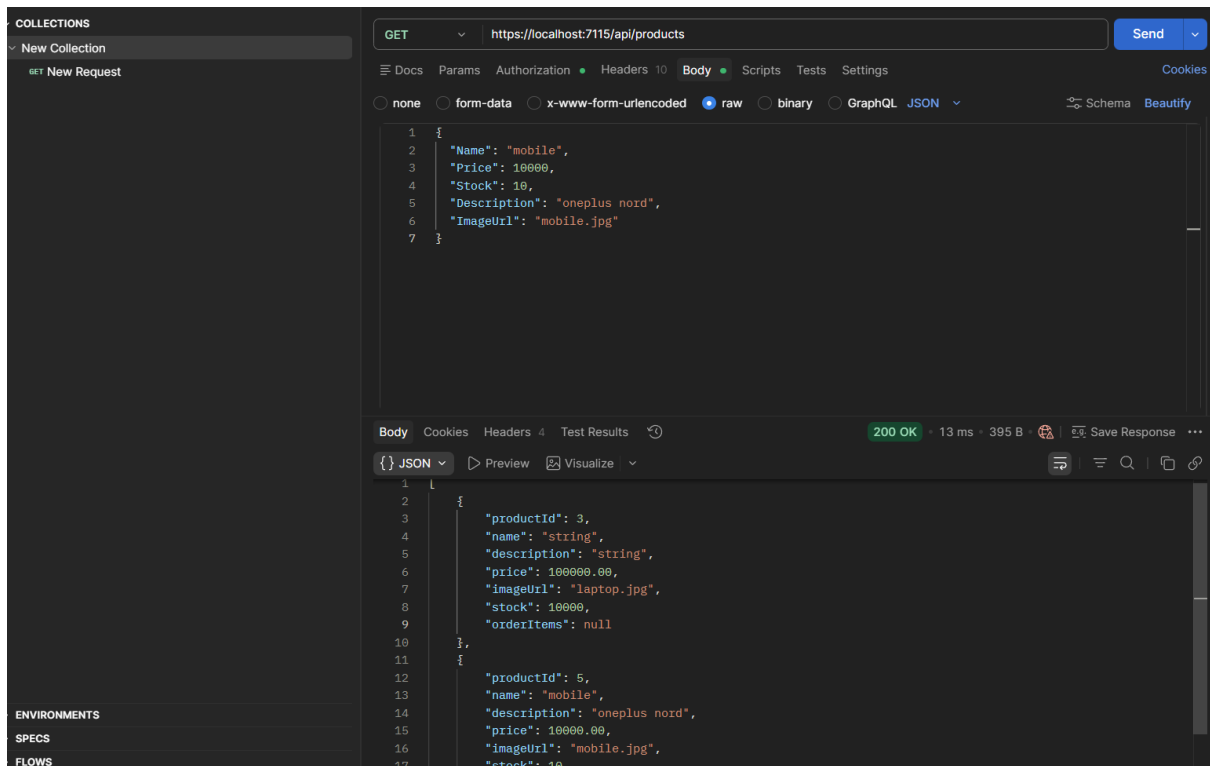


The screenshot shows the Postman interface for a POST request to `https://localhost:7115/api/products`. The request body is a JSON object:

```
1 {
2   "Name": "laptop",
3   "Price": 50000,
4   "Stock": 10,
5   "Description": "asus",
6   "ImageUrl": "laptop.jpg"
7 }
```

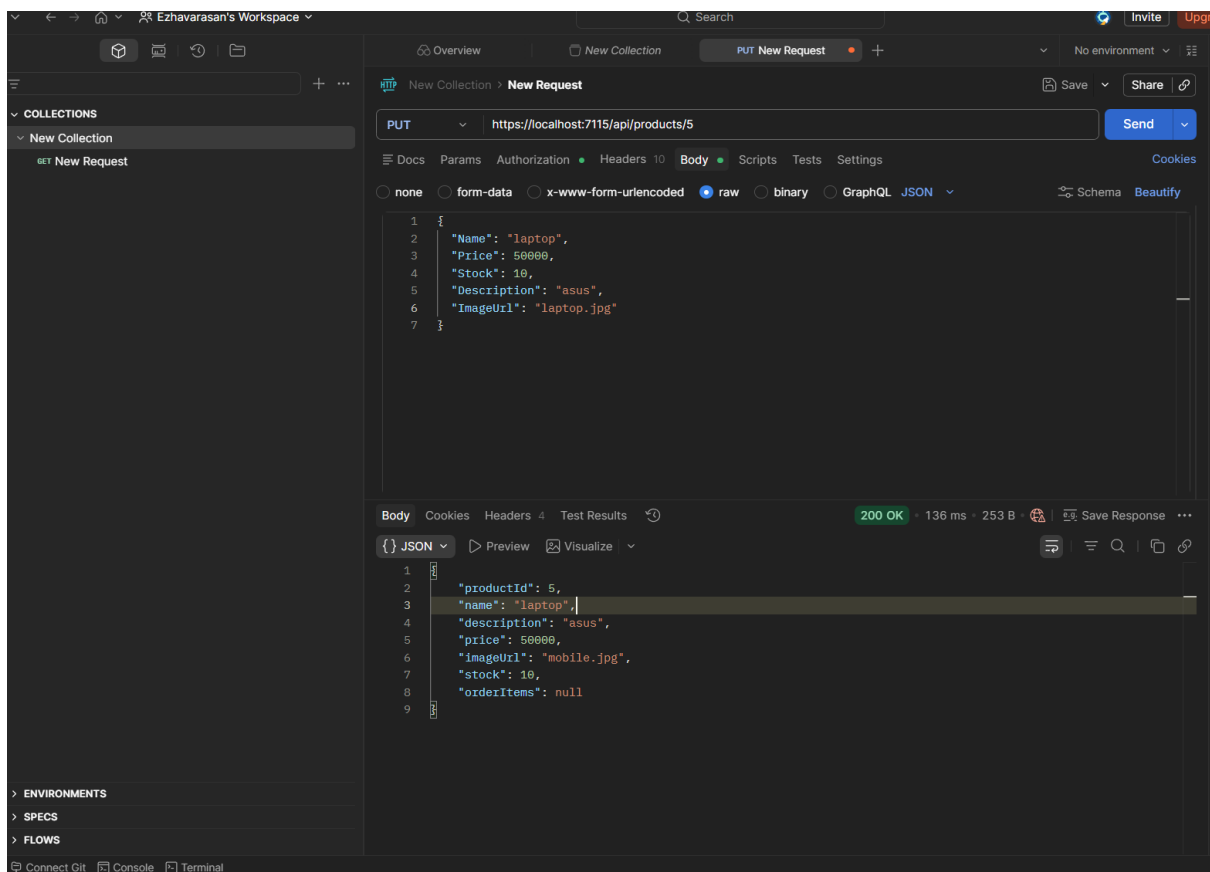
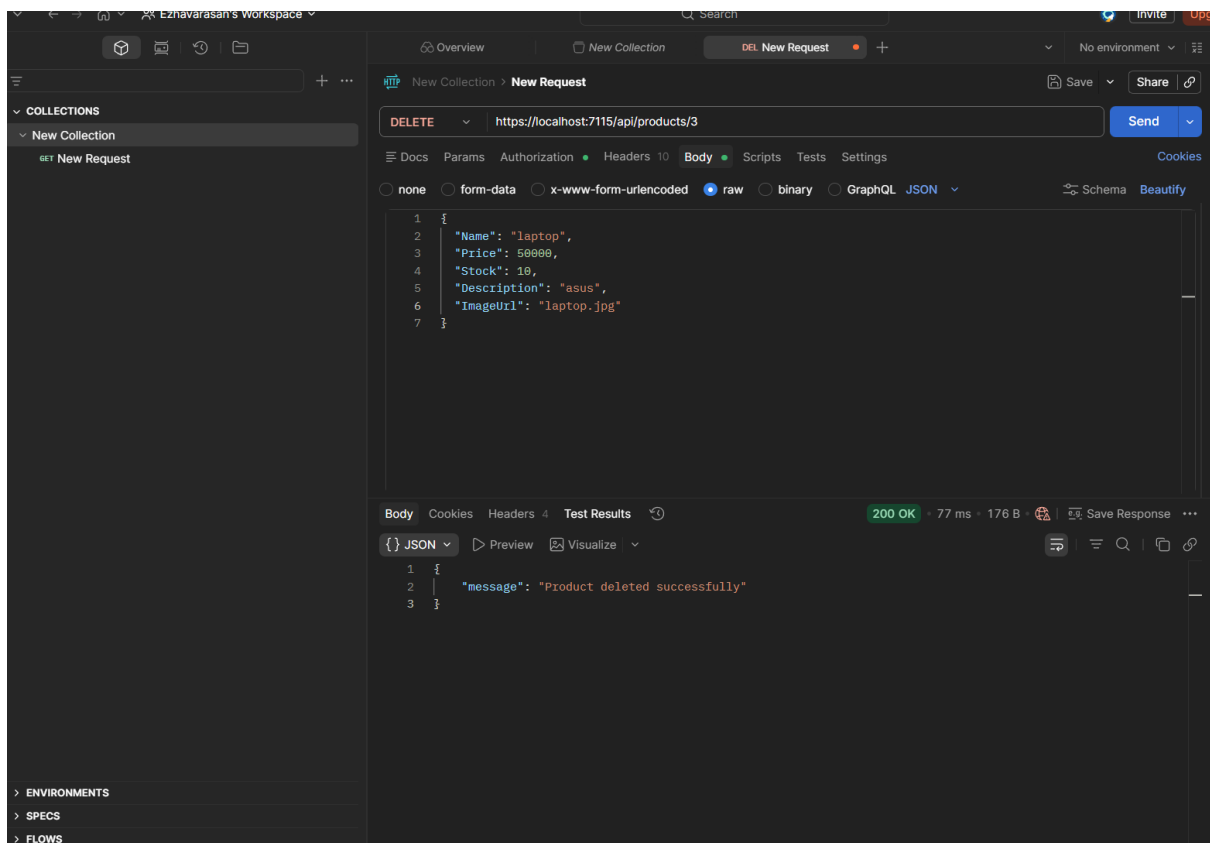
The response is `201 Created` with a status of `38 ms` and `307 B`. The response body is a JSON object:

```
1 {
2   "productid": 6,
3   "name": "laptop",
4   "description": "asus",
5   "price": 50000,
6   "imageUrl": "laptop.jpg",
7   "stock": 10,
8   "orderItems": null
9 }
```

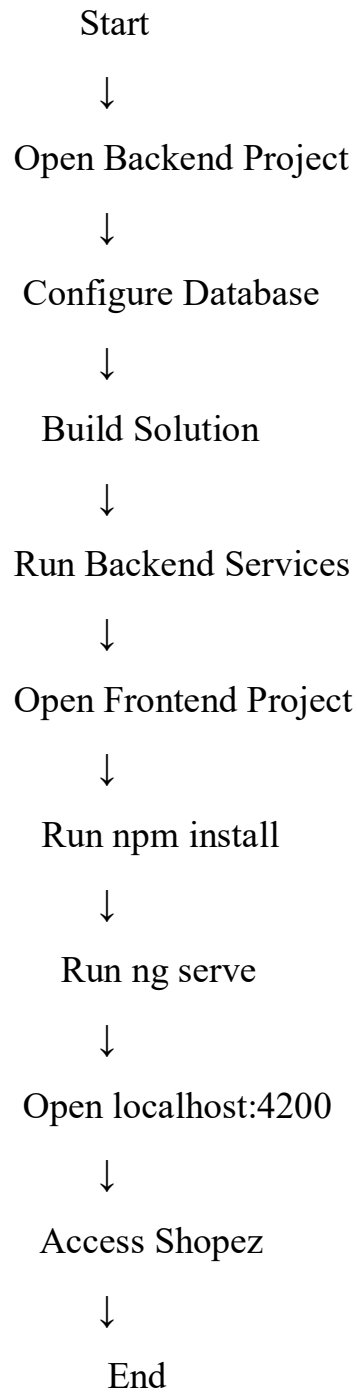


The screenshot shows the Postman interface for a GET request to `https://localhost:7115/api/products`. The response is `200 OK` with a status of `13 ms` and `395 B`. The response body is a JSON array of two product objects:

```
1 [
2   {
3     "productid": 3,
4     "name": "string",
5     "description": "string",
6     "price": 100000.00,
7     "imageUrl": "laptop.jpg",
8     "stock": 10000,
9     "orderItems": null
10  },
11  {
12    "productid": 5,
13    "name": "mobile",
14    "description": "oneplus nord",
15    "price": 10000.00,
16    "imageUrl": "mobile.jpg",
17    "stock": 10,
```



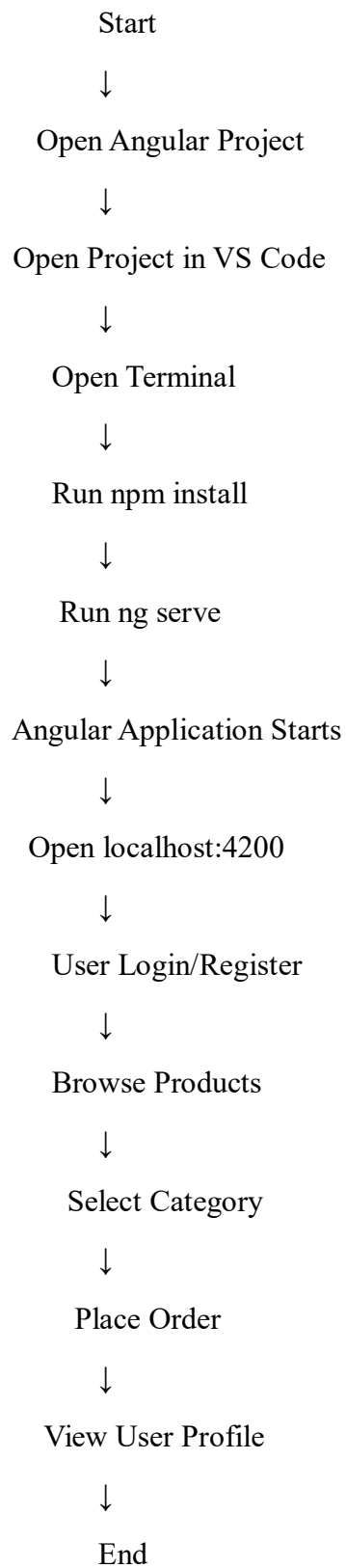
WorkFlow



Backend Workflow

1. Open the backend solution in Visual Studio.
2. Ensure SQL Server is running and database connection strings are configured properly in the application settings.
3. Set the required startup projects (User Service, Product Service, Order Service, and API Gateway) in Visual Studio.
4. Build the solution to restore dependencies and verify there are no errors.
5. Run the backend services by clicking Start or pressing F5 in Visual Studio.
6. Verify that the APIs are running successfully using Postman or browser Swagger endpoints.

WorkFlow



Frontend Workflow Description

The frontend workflow of Shopez begins by opening the Angular project in Visual Studio Code. Required dependencies are installed using `npm install`, and the application is started using `ng serve`. Once the Angular server runs successfully, users can access the application through the browser using `localhost:4200`. The user can then register or log in, browse products, view categories, place orders, and manage profile information through the application interface.

Future Enhancements

However, there are several enhancements that can be implemented in the future to improve the system's performance, scalability, security, and user experience.

One of the major future enhancements of Shopez is the implementation of a Payment Service to enable secure online transactions. Currently, the system focuses on product browsing, order placement, and management functionalities. In future versions, payment gateway integration can be added using secure payment platforms to support various payment methods such as credit/debit cards, UPI, net banking, digital wallets, and online banking services. This enhancement will provide customers with a complete end-to-end shopping experience by enabling secure and seamless online payments.

Another important enhancement is Cloud Deployment, which can significantly improve the availability, scalability, and performance of the Shopez application. Currently, the application runs in a local development environment; however, deploying the system on cloud platforms would enable global accessibility and better resource management. Cloud services can be used to host both frontend and backend applications, ensuring high availability, automatic scaling, and improved system performance during heavy traffic conditions. Cloud deployment would also support secure database management, backup facilities, monitoring, and disaster recovery mechanisms, making the application more robust and production-ready.

Additionally, future versions of Shopez may include features such as real-time order tracking, email and SMS notifications, AI-based product recommendations, wishlist functionality, advanced search and filtering, and multi-language support to further enhance the shopping experience. Security improvements such as multi-factor authentication and advanced monitoring systems can also be integrated to strengthen data protection.

Conclusion

The Shopez project has been successfully developed as a modern and scalable e-commerce web application to provide users with a smooth and secure online shopping experience. The application enables users to browse products, explore categories, place orders, and manage profile information efficiently through an easy-to-use interface. At the same time, administrators can effectively manage products, categories, and order-related operations through the admin dashboard.

The project was implemented using modern technologies such as Angular, ASP.NET Core Web API, SQL Server, and Microservices Architecture, ensuring better scalability, maintainability, and system performance. The integration of secure authentication mechanisms and API communication helped in building a reliable and efficient platform.

Through the development of Shopez, valuable practical knowledge and hands-on experience were gained in frontend and backend development, API integration, database management, authentication, and software architecture. The project also provided real-world exposure to problem-solving, system design, and full-stack application development.

Overall, Shopez successfully fulfills its objective of creating an efficient e-commerce platform and provides a strong foundation for future enhancements such as payment integration, cloud deployment, and advanced shopping features.

Declaration

I, **Ezhavarasan B**, hereby declare that the project titled “Shopez – E-Commerce Web Application” is an original work developed by me as part of practical learning and implementation in full-stack web development. This project has been completed under the valuable guidance and support of **Narasimha Sir**.

I further declare that the work presented in this project is genuine and has not been submitted previously for any other academic or professional purpose. All the efforts, implementation, and documentation have been carried out with sincerity and dedication to enhance technical knowledge and practical understanding of software development concepts.

I sincerely acknowledge the continuous support and learning opportunity provided by the **upGrad Team**, which contributed significantly to the successful completion of this project.